EXP 5

AIM : To simulate an LTE-Advanced wireless link using carrier aggregation and analyze the effect of aggregating multiple component carriers on available transmission bandwidth and estimated throughput using MATLAB.

Procedure:

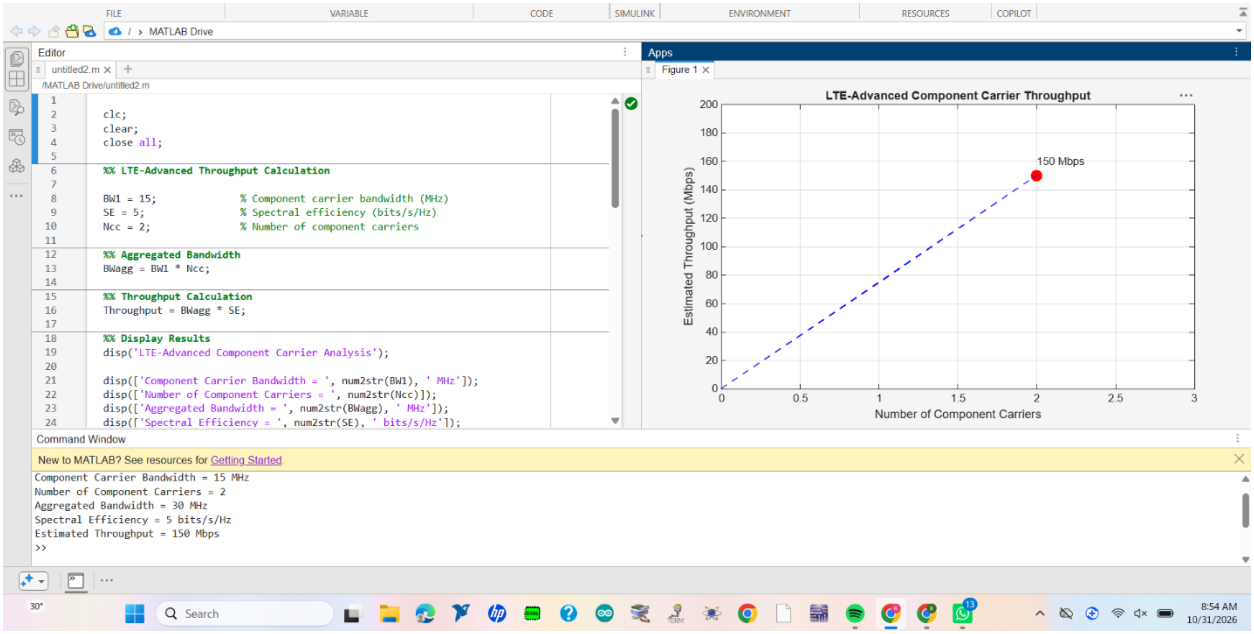
1. Open MATLAB and create a new script.
2. Define the component-carrier bandwidth, spectral efficiency and number of component carriers.
3. Calculate the single-carrier throughput using bandwidth × spectral efficiency.
4. Display the bandwidth and estimated throughput.
5. Plot the single-carrier result.

OBJ -1:CODE

```
clc;
clear;
close all;
BW1 = 15;
SE = 5;
Ncc = 2;
BWagg = BW1 * Ncc;
Throughput = BWagg * SE;
disp('LTE-Advanced Component Carrier Analysis');
disp(['Bandwidth = ', num2str(BWagg), ' MHz']);
disp(['Estimated Throughput = ', num2str(Throughput), ' Mbps']);
figure;
plot(Ncc, Throughput, 'ro', ...
```

```
    'MarkerSize', 10, 'MarkerFaceColor', 'r');
hold on;
plot([0 Ncc], [0 Throughput], 'b--', 'LineWidth', 1.5);
grid on;
xlabel('Number of Component Carriers');
ylabel('Estimated Throughput (Mbps)');
title('LTE-Advanced Throughput');
xlim([0 Ncc+1]);
ylim([0 Throughput+50]);
text(Ncc, Throughput+10, ...
    [num2str(Throughput), ' Mbps']);
hold off;
```

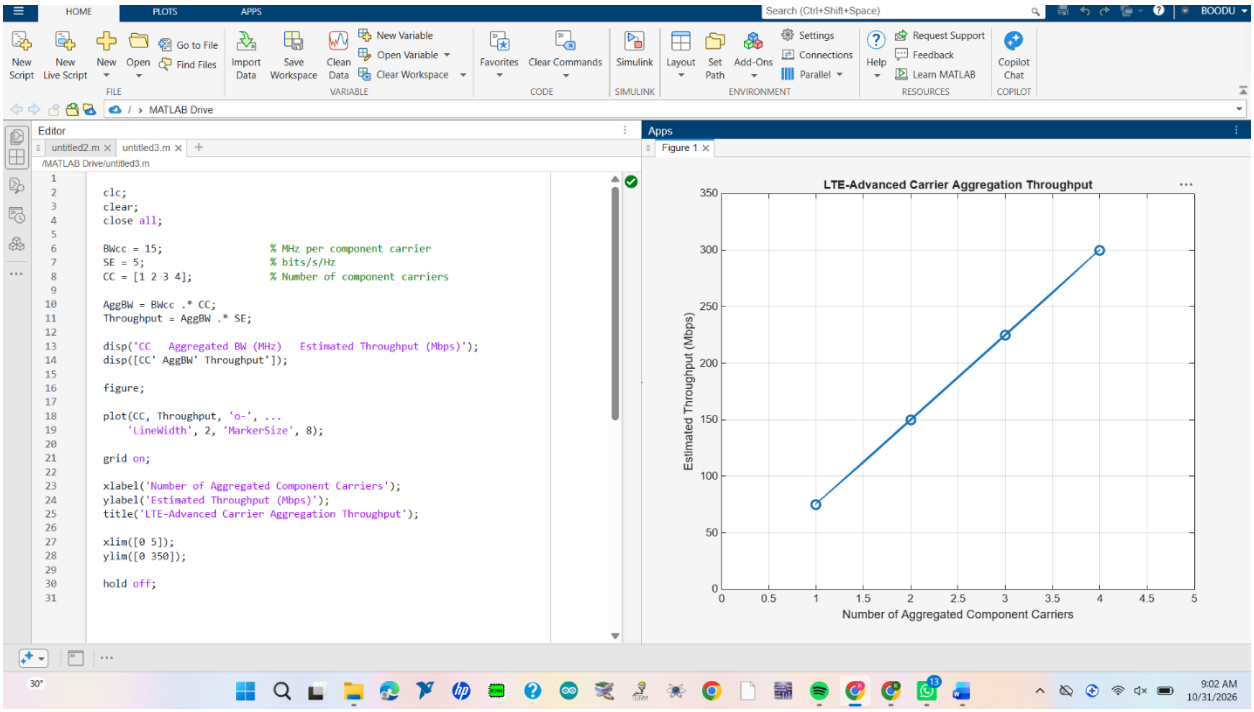
OUTPUT:



OBJ-2:CODE

```
clc;
clear;
close all;
BWcc = 15;                % MHz per component carrier
SE = 5;                   % bits/s/Hz
CC = [1 2 3 4];           % Number of component carriers
AggBW = BWcc .* CC;
Throughput = AggBW .* SE;
disp('CC   Aggregated BW (MHz)   Estimated Throughput (Mbps)');
disp([CC' AggBW' Throughput']);
figure;
plot(CC, Throughput, 'o-', ...
    'LineWidth', 2, 'MarkerSize', 8);
grid on;
xlabel('Number of Aggregated Component Carriers');
ylabel('Estimated Throughput (Mbps)');
title('LTE-Advanced Carrier Aggregation Throughput');
xlim([0 5]);
ylim([0 350]);
hold off;
```

OUTPUT:



OBJ-3:CODE

```
clc;
clear;
close all;
BW = [3 6 12 18 24];       % Carrier bandwidths (MHz)
SE = 5;                    % Spectral efficiency (bits/s/Hz)
TP = BW .* SE;             % Estimated throughput (Mbps)
disp('Bandwidth (MHz)   Throughput (Mbps)');
disp([BW' TP']);
figure;
plot(BW, TP, 'o-', 'LineWidth', 2, 'MarkerSize', 8);
```

```
grid on;
xlabel('Carrier Bandwidth (MHz)');
ylabel('Estimated Throughput (Mbps)');
title('Throughput vs Component Carrier Bandwidth');
xlim([0 26]);
ylim([0 140]);
```

OUTPUT:

